Analysis of the polynomial $x^3 + x^2 - 2x + 8$

In this worksheet we will study the field $K = \mathbb{Q}[\alpha] = \mathbb{Q}[X]/\langle P(X) \rangle$ where $P(X) = X^3 + X^2 - 2X + 8$.

Routine algebraic calculations will be carried out using the Python package `sympy`.

```
from sympy import var, discriminant, resultant,\
 factor, factorint, expand, div
```

```
var('x')
pol = x**3 + x**2 - 2*x + 8
pol
```

$x^3 + x^2 - 2x + 8$

For various calculations we will need the derivative of $P(X)$ with respect to X.

```
dpol=pol.diff(x)
dpol
```

$3x^2 + 2x - 2$

In order to study the ring $\mathbb{Z}[\alpha]$ we will need the discriminant of $P(X)$ with respect to X.

```
discriminant(pol,x)
```

-2012

Since the discriminant is negative, there is one real root and two complex roots.

We find the prime factors of the discriminant to know at what primes p we *could* the ring of integers to be *larger* than $\mathbb{Z}_{(p)}[\alpha]$.

```
factorint(2012)
```

```
{2: 2, 503: 1}
```

Thus, we have $(-2012) = (-1) \cdot 2^2 \cdot 503$.

Analysis for $p = 503$

Since $P(X)$ has a multiple root modulo $p = 503$, it has a common root with $P'(X)$ in $\mathbb{F}_{503}$. We wish to find this root.

To do so, we use the GCD algorithm to calculate a common factor of $P(X)$ and $P'(X)$ in $\mathbb{F}_{503}$.

To use the GCD algorithm we first replace $P'(X)$ by a multiple that is monic in $\mathbb{F}_{503}$.

```
dpol*(504//3)
```

$504x^2 + 336x - 336$

Thus, in $\mathbb{F}_{503}$ we can use $P_1(X) = X^2 + 336X - 336$ in place of $P'(X)$.

```
dpol1=x**2+336*x-336
```

The remainder on division of $P(X)$ by $P_1(X)$ *must* be the common linear factor.

```
lin=div(pol,dpol1)[1]
lin=(lin.coeff(x)%503)*x+lin.coeff(x,n=0)%503
lin
```

$222x + 120$

Make this monic in $\mathbb{F}_{503}$

```
a=pow(lin.coeff(x,n=1),503-2,503)
lin=a*lin
lin=(lin.coeff(x,n=1)%503)*x+lin.subs({x:0})%503
lin
```

$x + 354$

We now find the remaining linear factor of $P(X)$ over $\mathbb{F}_{503}$.

```
olin=div(pol,lin**2)[0]
b=pow(olin.coeff(x,n=1),503-2,503)
olin=b*olin
olin=(olin.coeff(x,n=1)%503)*x+olin.subs({x:0})%503
olin
```

$x + 299$

This gives us the two roots of $P(X)$ in $\mathbb{F}_{503}$.

```
(503-olin.coeff(x,n=0), 503-lin.coeff(x,n=0))
```

```
(204, 149)
```

Here 204 is a simple root and 149 is a double root.

Just a quick check that $(X - 149)^2 \cdot (X - 204)$ equals $P(X)$ modulo 503.

```
check=expand(pol-(x-149)**2*(x-204))
[check.coeff(x**i)%503 for i in [0,1,2]]
```

```
[0, 0, 0]
```

Since 204 is a simple root, we can use Hensel lifting to lift it to a root η in $\mathbb{Z}_{503}$.

Thus, $P(X)$ becomes $(X - \eta)Q(X)$ in $\mathbb{Z}_{503}[X]$ for some quadratic polynomial $Q(X)$.

We will prove that the image of $Q(X)$ is irreducible in $\mathbb{Z}[X]/\langle 503^2 \rangle$. It will follow that $Q(X)$ is irreducible in $\mathbb{Z}_{503}[X]$. Now $Q(X) = (X - 149)^2$ in $\mathbb{F}_{503}[X]$. Thus, the ring $R = \mathbb{Z}_{503}[X]/\langle Q(X) \rangle$ is a ramified extension of degree 2 of $\mathbb{Z}_{503}$.

It follows that
$$\frac{\mathbb{Z}_{503}[X]}{\langle P(X)\rangle} = \mathbb{Z}_{503} \times R$$

Hensel lifting the simple root

We first calculate η modulo 503^2.

To do this we need to calculate the inverse of $P'(204)$ in $\mathbb{F}_{503}$.

```
dpol.subs({x:204})%503
```

7

```
7*72
```

```
504
```

Thus, 72 is the required inverse which we can use to calculate the Hensel lift.

```
rt2=(204-72*pol.subs({x:204}))%(503*503)
rt2
```

82696

Check that this is a root!

```
pol.subs({x:rt2})%(503*503)
```

0

We divide $P(X)$ by $(X - 82696)$ to get the quadratic polynomial $Q(X)$ modulo 503^2.

```
qpol=div(pol,x-rt2)[0]
qpol
```

$x^2 + 82697x + 6838711110$

We first note that $Q(X) - (X - 149)^2$ is diviible by 503.

```
rqpol=expand(qpol-(x-149)**2)
rqpol.coeff(x,n=1)%503,rqpol.coeff(x,n=0)%503
```

```
(0, 0)
```

We then use the Eisenstein criterion to prove that $Q(X)$ is irreducible in $\mathbb{Z}[X]/\langle 503^2\rangle$.

```
sqpol=expand(qpol.subs({x:x+149}))
sqpol.coeff(x,n=1)%503,sqpol.coeff(x,n=0)%(503**2)
```

```
(0, 77462)
```

Since $Q(X)$ in $\mathbb{Z}_{503}[X]$ is a lift of this polynomial, it is also irreducible.

As seen above this means that $\mathbb{Z}_{503}[\alpha]$ is of the form $\mathbb{Z}_{503} \times R$ where R is a ramified extension of degre 2 of $\mathbb{Z}_{503}$.

Analysis at $p = 2$

We note that modulo $p = 2$, the polynomial $P(X)$ becomes $X^3+X^2 = X^2(X+1)$.

Hence, there is a simple root $X = 1$ in $\mathbb{F}_2$ which can be lifted as above.

```
dpol.subs({x:1})%2
```

1

```
rt2 = (1 - pol.subs(x,1))%4
rt3 = (rt2 - pol.subs(x,rt2))%8
rt4 = (rt3 - pol.subs(x,rt3))%16
rt2,rt3,rt4
```

```
(1, 1, 9)
```

We can continue this process to obtain a lift ξ in $\mathbb{Z}_2$ of the root of $P(X)$. We then write $P(X) = (X - \xi)Q(X)$ for a quadratic polynomial in $\mathbb{Z}_2[X]$.

We can calculate the image of $Q(X)$ in $\mathbb{Z}[X]/\langle 16\rangle$.

```
qpol=div(pol,x-rt4)[0]
qpol
```

$x^2 + 10x + 88$

We note that this is *not* an Eisenstein polynomial!

In fact, it has distinct roots in $\mathbb{Z}/\langle 16\rangle$ as we shall see below.

Alternate Hensel lifting

To study the roots of $P(X)$ which are divisible by 2 in $\mathbb{Z}_2$ we substitute X by $2X$ and clear denominators to obtain a new polynomial $F(X)$. We then check for roots of $F(X)$ in $\mathbb{Z}_2$.

```
fpol = div(pol.subs({x:2*x}),4)[0]
fpol
```

$2x^3 + x^2 - x + 2$

We also calculate its derivative $F'(X)$.

```
dfpol=fpol.diff(x)
dfpol
```

$6x^2 + 2x - 1$

Note that $X = 0$ is a root of $F(X)$ and is not a root of $F'(X)$ in $\mathbb{F}_2[X]$.

Similarly $X = 1$ is a root of $F(X)$ and is not a root of $F'(X)$ in $\mathbb{F}_2$.

```
frt=[0,1]
for i in frt:
    print(fpol.subs({x:i})%2,dfpol.subs({x:i})%2)
```

```
0 1
0 1
```

We can use the stronger version of Hensel's lemma to lift these roots to $\mathbb{Z}_2$.

```
frt2 = [(frt[i]+fpol.subs(x,frt[i]))%4 for i in [0,1]]
frt3 = [(frt2[i]+fpol.subs(x,frt2[i]))%8 for i in [0,1]]
frt4 = [(frt3[i]+fpol.subs(x,frt3[i]))%16 for i in [0,1]]
frt2,frt3,frt4
```

```
([2, 1], [6, 5], [6, 5])
```

We can check that these are roots modulo 16.

```
for i in frt4:
    print(fpol.subs({x:i})%16)
```

```
0
0
```

This lifting gives us two roots μ and ν of $F(X)$ in $\mathbb{Z}_2$.

It follows that 2μ and 2ν are roots of $P(X)$ in $\mathbb{Z}_2$.

In particular, we see that the ring of integers of $\mathbb{Z}_2 \otimes_{\mathbb{Z}} \mathcal{O}_K$ contains elements that are *not* in $\mathbb{Z}_2[\alpha]$.